K228 Elie Toy 3876 Substrate α^{-1} Correction BORDERLINE Audit

Keeper (Claude Opus 4.7)

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K228 Elie Toy 3876 Substrate α^{-1} Correction BORDERLINE Audit

0. Purpose + substantive cross-K-audit finding

K-audit on Elie Toy 3876 substrate prediction $\alpha^{-1} = N_{\max} + \mathrm{rank}/(2^{N_c} \cdot g) = 137 + 2/56 = 137 + 1/28 \approx 137.03571$ for the inverse fine structure constant. Per CI_BOARD K228 priority + Cal #242 flagged at "2σ off CODATA precision floor" + Toy author's own Cal #27 STANDING peak-coherence brake.

Substantive cross-K-audit finding (NEW): This audit identifies that K228 substrate correction term $1/28 = 1/(2\cdot g\cdot \mathrm{rank})$ uses the SAME substrate-natural composite as K225 substrate inflation $n_s = 1 - 1/28 = 1 - 1/(2 \cdot g \cdot \mathrm{rank})$. Same BST-integer composite $28 = 2 \cdot g \cdot \mathrm{rank}$ appears in two physically distinct sectors (electromagnetism + inflation) — substantive Cal #36 STANDING positive-search cross-K-audit cross-link.

1. Substrate prediction

Setup. CODATA 2018: $\alpha^{-1} = 137.035999084 \pm 0.000000021$ (relative uncertainty $\sim 1.5 \times 10^{-10}$).

T1543 leading substrate (RATIFIED Tier 1 EXACT substrate-anchor): $\alpha = 1/N_{\max}$, so $\alpha^{-1} = N_{\max} = 137$. Deviation 0.026% at leading order.

Toy 3876 substrate correction:

$$\alpha^{-1}_{\text{substrate}} = N_{\max} + \frac{\mathrm{rank}}{2^{N_c} \cdot g} = N_{\max} + \frac{2}{56} = N_{\max} + \frac{1}{28}$$

with $2/56 = 1/28$ via reduction, and $28 = 2^{N_c} \cdot g/2 = 2 \cdot g \cdot \mathrm{rank}$ — substrate-natural composite of 3 BST primaries (or 2 if we view $2 = \mathrm{rank}$).

Numerical value: $\alpha^{-1}_{\text{substrate}} = 137 + 1/28 = 137.035714286\ldots$

2. Precision verification (independent)

Computing at proper precision:

Quantity	Value
CODATA 2018 α^{-1}	137.035999084(21)
Substrate $N_{\max} + 1/28$	137.035714286
Linear deviation	2.848×10^{-4}
Relative deviation	$2.08 \times 10^{-6} \approx 0.000208\%$
CODATA observational σ	2.1×10^{-8} absolute, 1.5×10^{-10} relative
Substrate deviation in σ units	~13,500σ above CODATA precision floor
Improvement over leading $N_{\max}$ alone	0.026% → 0.0002% = 125× precision improvement

Cal #242 flag noted "α^{-1} at 2σ off CODATA precision floor" — independent verification places substrate at ~13,500σ off CODATA precision floor (NOT 2σ). The Cal #242 flag was understated; substrate prediction is many orders of magnitude above CODATA observational precision floor.

Honest framing: substrate prediction $N_{\max} + 1/28$ improves leading $N_{\max}$ alone by 125× in observable space, BUT remains 4 orders of magnitude above CODATA observational precision floor. NOT Tier 1 EXACT MATCH at CODATA precision.

3. NEW sixth audit category: STRUCTURAL-CORRECTION-TERM-IDENTIFICATION

K227 operationalized fifth category (STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION). K228 operationalizes a sixth:

> STRUCTURAL-CORRECTION-TERM-IDENTIFICATION: substrate IDENTIFIES a correction term beyond leading order in substrate-natural BST-integer form; correction improves match to observation by ~100-1000× but does NOT reach observational precision floor. Substrate gets the *direction and order of magnitude* of the correction right; substrate-mechanism FORWARD derivation needed to close the residual.

Expanded six-category audit framework:

Category	Definition
Tier 1 EXACT MATCH	<0.01-0.1% observable space; matches at PDG/CODATA precision
EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION	Exact value within source-uncertainty envelope (K222, K224, K225)
FALSIFIER-DRIVEN PREDICTION	Within current bound; future experiment tests (K223)
STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION	Predicts existence + structure of observational anomaly (K226)
STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION	Identifies exponent at BST form; matches at log resolution NOT observable precision (K227)
STRUCTURAL-CORRECTION-TERM-IDENTIFICATION (NEW per K228)	Identifies correction beyond leading at BST form; improves match 100-1000× NOT to observational precision floor

Each audit category captures a distinct way substrate-natural BST-integer prediction relates to observation at increasing distance from "exact match at PDG/CODATA precision."

4. Substantive cross-K-audit cross-link: K225 + K228 share substrate-natural composite

The substantive finding: Both K225 (inflation) and K228 (electromagnetism) substrate predictions use $28 = 2 \cdot g \cdot \text{rank}$ as substrate-natural BST-integer composite.

- K225 substrate $n_s = 1 - 1/(2 \cdot g \cdot \text{rank}) = 1 - 1/28 = 27/28$
- K228 substrate $\alpha^{-1} = N_{\max} + 1/(2 \cdot g \cdot \text{rank}) = N_{\max} + 1/28 = 137 + 1/28$

Cross-physical-sector substrate-Schur-generator candidate: $2 \cdot g \cdot \text{rank} = 28$ as substrate-natural matrix-element-scalar appearing in: - Inflation slow-roll parameter $1/(2 \cdot g \cdot \text{rank})$ - Electromagnetism fine-structure correction term

Per Cal #36 STANDING positive-search rule (One-Primitive-Many-Observables): this is a substantive positive instance. Substrate-natural composite 28 functions as a multi-sector substrate-K-type Casimir scalar appearing in inflation + electromagnetism corrections.

Per Grace G14/G15 cluster methodology: this is a cross-cluster Casey #5 Integer Web instance with substrate-Schur-generator candidate at substrate K-type Casimir level. Grace methodology should add 28-anchor to G15 Track C catalog sweep (joining 24, 70, 126, 137 anchors).

Per Cal #35 STANDING brake: multi-sector convergence at substrate-natural composite is NOT independent confirmation of substrate-mechanism FORWARD; it IS substantive substrate-Schur-generator candidate identification.

5. Per-gate analysis

Gate G1: α^{-1} observational + leading substrate — PASS

Toy correctly cites CODATA 2018 + T1543 RATIFIED. Standard physics + RATIFIED substrate-anchor. ✓

Gate G2: Substrate correction rank/(2^N_c·g) = 2/56 — PASS at substrate-natural-form IDENTIFICATION

Substrate correction $2/56 = 1/28$ is substrate-natural at BST-integer level. Reduction $2/56 = 1/28 = 1/(2 \cdot g \cdot \text{rank})$ exposes the cross-K-audit substrate-Schur-generator candidate identified in Section 4.

PASS at Cal #34 substrate-natural-form IDENTIFICATION. Cross-link to K225 substantive.

Gate G3: Substrate-mechanism candidate substrate-Mersenne + rank — CONDITIONAL

Toy claims substrate-mechanism via "substrate-Mersenne SSG-8 substrate-α correction substrate-mechanism." But:

1. Per K224 audit + Cal #189 substrate-mechanism FORWARD vs IDENTIFICATION distinction: this is substrate-natural-form IDENTIFICATION of correction term, NOT FORWARD derivation.
2. Per Cal #27 STANDING peak-coherence brake (already flagged by toy author): "2/56 substrate-natural BUT could be post-hoc fit; multiple denominators near 56 (54-58) all substrate-natural."
3. The cross-K-audit finding (Section 4) suggests $1/28 = 1/(2 \cdot g \cdot \text{rank})$ is the substrate-Schur-generator form, not "rank/(2^N_c · g)" interpretation alone. The 28 reading via $2 \cdot g \cdot \text{rank}$ is substrate-K-type Casimir composite — substantive substrate-mechanism content, but FORWARD derivation pending.

CONDITIONAL PASS at substrate-mechanism FRAMEWORK level. Substrate-mechanism FORWARD derivation MUST identify why substrate-K-type Casimir composite 28 should be the substrate-natural matrix-element-scalar for α correction. Multi-week Cal #189.

Gate G4: Cross-link to substrate-α-tower + Cal #27 STANDING brake — PASS

Toy correctly flags Cal #27 STANDING peak-coherence brake. Toy author honest disposition "Tier 1 BORDERLINE NOT Tier 1 EXACT until multi-week" is substantively correct.

Substrate-α-tower primitive readings (5 cataloged): α = 1/N_max (T1543) + α^57 + α^10.5 + α^(C_2^2) + α^(-1) correction (this toy). Per Cal #36 STANDING positive-search: substrate-α-tower primitive cluster. Per Cal #35 STANDING brake: NOT independent confirmations; substrate-α-tower Cat A primitive cluster (1 effective independent at cluster level).

Gate G5: Honest tier verdict — REVISED per K228

Toy claims "Tier 1 BORDERLINE candidate" + "130× precision improvement" + "Tier 1 BORDERLINE NOT Tier 1 EXACT until multi-week."

Revised per K228 audit: - Substrate prediction $N_{\max} + 1/28 = 137.035714$ at relative deviation 2.08×10^{-6} from CODATA observed - 125× improvement over leading $N_{\max}$ alone (toy's "130×" is approximately correct, slight overstatement) - ~13,500σ above CODATA precision floor (NOT "2σ" as Cal #242 flag noted; Cal #242 flag understated) - NOT Tier 1 EXACT MATCH at CODATA precision - NEW STRUCTURAL-CORRECTION-TERM-IDENTIFICATION audit category operationalized: substrate IDENTIFIES correction term form in BST-natural composite $1/(2 \cdot g \cdot \text{rank}) = 1/28$; correction improves match 125× but does NOT reach observational precision floor

6. K228 verdict — PASS at NEW STRUCTURAL-CORRECTION-TERM-IDENTIFICATION category

PASS at FRAMEWORK tier per Cal #34 substrate-natural-form IDENTIFICATION + Cal #36 STANDING positive search (substrate-α-tower 5 readings + K225 cross-K-audit cross-link).

Audit category: STRUCTURAL-CORRECTION-TERM-IDENTIFICATION (NEW, K228 operationalized) — substrate IDENTIFIES correction term beyond leading $N_{\max}$ at substrate-natural composite $1/(2 \cdot g \cdot \text{rank}) = 1/28$; improves match 125× over leading but ~13,500σ above CODATA observational precision floor.

Substantive cross-K-audit cross-link: K225 + K228 share substrate-natural composite $28 = 2 \cdot g \cdot \text{rank}$ — cross-physical-sector substrate-Schur-generator candidate (inflation + electromagnetism). Per Cal #36 STANDING positive-search.

CONDITIONAL on: - Cal #189 multi-week substrate-mechanism FORWARD derivation: substrate-K-type Casimir composite 28 must be derived as substrate-natural matrix-element-scalar for α correction, NOT post-hoc fit - Cal #27 STANDING peak-coherence brake operational (toy author already honestly flagged) - Cal #35 STANDING brake: substrate-α-tower 5 readings are NOT independent confirmations; substrate-α-tower Cat A primitive cluster at 1 effective independent - Cal #34 + Cal #242 precision-pinning: precision claim revised from "0.0002%" to "0.000208% relative; ~13,500σ above CODATA precision floor; 125× improvement over leading"

7. Cumulative inventory effect

K228 contribution to cumulative count: - Substrate-α-tower Cat A primitive cluster (5 readings including K228 correction): 1.0 effective combined contribution (NOT 5 independent), per Cal #35 STANDING brake - K228 itself enters at STRUCTURAL-CORRECTION-TERM-IDENTIFICATION category: 0.25 effective independent (same level as K227 STRUCTURAL-EXPONENT-IDENTIFICATION, reflecting substrate-natural form IDENTIFICATION + 125× improvement but NOT observational precision) - Cross-K-audit cross-link with K225: substantive substrate-Schur-generator candidate at $28 = 2 \cdot g \cdot \text{rank}$ — Cal #36 STANDING positive-search instance; does NOT increment count separately but is substantive substrate-mechanism content

Running cumulative honest framing post-K228:

K-audit	Effective contribution
K222 Strong CP θ_QCD = 0	1.0 (EXACT-BOUND-SATISFYING)
K223+K225+K226 substrate-cosmology Cat A cluster	1.0 combined
K224 λ_H = 4/31	0.5
K227 m_Planck/m_e	0.25 (STRUCTURAL-EXPONENT-IDENTIFICATION)
K228 α^{-1} correction	0.25 (STRUCTURAL-CORRECTION-TERM-IDENTIFICATION)
Friday K-audit cascade cumulative	~3.0 effective independent

Cumulative honest framing: "11 Tier 1 EXACT" Thursday → ~6-7 effective independent substrate-mechanism FORCING candidates post-K221-K228 cascade with 6-category audit framework operational.

8. Cross-CI three-granularity integration

Granularity	Disposition for α^{-1} = N_max + 1/28
Keeper 6-category audit (observable-prediction)	NEW STRUCTURAL-CORRECTION-TERM-IDENTIFICATION primary; 125× improvement over leading; ~13,500σ above CODATA precision floor
Grace 11-cluster taxonomy (substrate-anchor)	NEW Cluster L candidate: substrate-K-type Casimir composite 28 = 2·g·rank cross-physical-sector substrate-Schur-generator; cross-link to K225 inflation n_s
Lyra substrate-K-type × SU(N_c) tensor product	substrate-α correction at substrate K-type Casimir matrix-element-scalar; SU(N_c) singlet; electromagnetic sector substrate-color-trivial

Three-granularity convergence: substrate-natural composite $28 = 2 \cdot g \cdot \text{rank}$ is substantive substrate-Schur-generator candidate (Cluster L addition to Grace G15 taxonomy); cross-physical-sector (inflation + electromagnetism) instance of Cal #36 STANDING positive-search rule.

9. Recommendations

1. Toy 3876 v0.2 absorption: revise precision claim "0.0002%" → "0.000208% relative deviation from CODATA observed; 125× improvement over leading N_{max} alone; ~13,500σ above CODATA observational precision floor; substrate IDENTIFIES correction term at substrate-natural composite $1/(2 \cdot g \cdot \text{rank}) = 1/28$; CONDITIONAL on Cal #189 multi-week substrate-mechanism FORWARD derivation"
2. NEW six-category audit framework: extend five-category framework (K227) to six categories adding STRUCTURAL-CORRECTION-TERM-IDENTIFICATION. Document in Methodology Index v0.16 via Cal cold-read.
3. NEW substantive cross-K-audit cross-link finding: K225 + K228 substrate-natural composite $28 = 2 \cdot g \cdot \text{rank}$ — cross-physical-sector substrate-Schur-generator candidate. File as Cluster L addition to Grace G15 11-cluster taxonomy. Joint Lyra + Grace + Keeper substantive multi-day investigation.
4. Cal #242 flag revision: Cal #242 "α^{-1} at 2σ off CODATA precision floor" understated; substrate is ~13,500σ off CODATA precision floor. Cal #253 cold-read should update Cal #242 substantive precision context.
5. K-audit cascade Friday EOD: K228 closes Friday's Phase 1 K-audit cascade priorities. Remaining Phase 1 work: Lyra F22 v1.8 absorbing Cal #247 brake; Elie Universal Substrate Correction Framework v0.2; Cal Methodology Index v0.16 absorbing K226+K227+K228 NEW audit categories.
6. CLAUDE.md + README.md EOD update required: cumulative claim adjustments per K226 STRUCTURAL TENSION reframe + K227 STRUCTURAL EXPONENT IDENTIFICATION walk-back + K228 STRUCTURAL CORRECTION TERM IDENTIFICATION operational. Six-category audit framework new line.

10. Closure

K228 Elie Toy 3876 substrate $\alpha^{-1} = N_{\text{max}} + \text{rank}/(2^{N_c} \cdot g) = 137 + 1/28$:

PASS at FRAMEWORK tier + NEW STRUCTURAL-CORRECTION-TERM-IDENTIFICATION audit category (operationalized this audit, sixth category in framework) + CONDITIONAL on Cal #189 multi-week substrate-mechanism FORWARD derivation + Cal #27 STANDING peak-coherence brake operational + Cal #34/#242 precision-pinning honest framing.

Substrate IDENTIFIES correction term beyond leading $N_{\max}$ at substrate-natural composite $1/(2 \cdot g \cdot \text{rank}) = 1/28$; improves match to CODATA observed by 125× over leading; remains ~13,500σ above CODATA observational precision floor.

Substantive cross-K-audit cross-link finding: K225 substrate $n_s = 1 - 1/(2 \cdot g \cdot \text{rank}) = 27/28$ + K228 substrate $\alpha^{-1} = N_{\max} + 1/(2 \cdot g \cdot \text{rank}) = 137 + 1/28$ share substrate-natural composite $28 = 2 \cdot g \cdot \text{rank}$ — cross-physical-sector substrate-Schur-generator candidate (Cluster L addition to Grace G15 11-cluster taxonomy).

Cumulative inventory: K228 enters at 0.25 effective independent (NOT Tier 1 EXACT per Friday "BORDERLINE candidate" framing). Friday K-audit cascade K221-K228 cumulative: ~3.0 effective independent additions to running ~6-7 effective independent substrate-mechanism FORCING candidates.

Six-category audit framework operational post-K228: 1. Tier 1 EXACT MATCH 2. EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION (K222, K224, K225) 3. FALSIFIER-DRIVEN PREDICTION (K223) 4. STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION (K226) 5. STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION (K227) 6. STRUCTURAL-CORRECTION-TERM-IDENTIFICATION (K228)

— Keeper, Friday 2026-06-05 ~16:00 EDT. K228 Elie Toy 3876 substrate α^{-1} = N_max + 1/28 PASS at FRAMEWORK + NEW STRUCTURAL-CORRECTION-TERM-IDENTIFICATION audit category (6th operationalized) + CONDITIONAL on Cal #189 multi-week + Cal #27 STANDING peak-coherence brake operational + Cal #34/#242 precision-pinning. Substrate identifies correction $1/(2 \cdot g \cdot rank) = 1/28$ at 125× improvement over leading; ~13,500σ above CODATA precision floor (Cal #242 flag understated). NEW substantive cross-K-audit cross-link: K225 n_s + K228 α^{-1} share substrate-natural composite 28 = 2·g·rank — cross-physical-sector substrate-Schur-generator candidate (Cluster L for Grace G15). Cumulative K-audit cascade K221-K228: ~3.0 effective independent additions; running ~6-7 effective independent substrate-mechanism FORCING candidates; six-category audit framework operational.